

Short Report for Book-Swap Application

Name and Surname

- **Name:** Vanja Antonovic
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URL to Git Repository

- **Repository URL:** <https://github.com/iSoulFeed1g/Book-Swap-Local>
(I fixed a few bugs and added a few QOL features since last year)
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URL to Demo Video

- **Demo Video URL:** <https://www.youtube.com/watch?v=3T97mrREIVk>

Note: The video doesn't include labels. I couldn't get the editing app to work, I also didn't showcase the search functionality. This is only a demo video I will showcase the finished website when I have the oral exam

URL to Application

- **Application URL:** <https://www.studenti.famnit.upr.si/~89211069/Book-Swap-Local/>
 - **Application URL2:** <https://isoulfeed1g.github.io/Book-Swap-Local/>
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Functionalities Overview

1. User Registration and Authentication

Brief Description: The project includes a user registration system that allows new users to sign up by providing their name, email, and password. Once registered, users can log in, and the system authenticates their credentials to grant them access to specific features.

Most Difficult Part: The most challenging aspect was implementing a system that automatically assigns a default profile picture to new users upon registration. This required careful handling of file paths and ensuring that the default image was consistently applied across different parts of the application without disrupting the user experience.

2. Book Listing and Management

Brief Description: This feature allows users to create, edit, and delete book listings on the platform. Users can provide details such as the book's title, author, genre, price, and a brief description. Additionally, users have the ability to manage their listings directly from their profile.

Most Difficult Part: The biggest challenge in this feature was handling the dynamic updating of book listings. Implementing the functionality to allow users to edit and delete listings required careful management of database operations to ensure data consistency and integrity, especially when multiple users might be interacting with the same data simultaneously.

3. Search and Filter Functionality

Brief Description: The project includes a search bar that lets users search for books by title and author. Additionally, users can filter search results based on criteria like price, genre, and condition, making it easier to find specific books.

Most Difficult Part: The most challenging part of this feature was implementing an intuitive user interface that allowed users to easily combine multiple filters without confusion. Ensuring that the user experience remained smooth and that the filters interacted correctly with one another took several iterations of the design. Additionally, managing the potential combinations of filters and ensuring accurate results were returned required careful logic and testing.

4. Messaging System

Brief Description: The messaging system allows users to communicate directly with sellers through a chat interface. It also includes a notification system that displays a red dot on the inbox icon when there are unread messages.

Most Difficult Part: Integrating real-time messaging with notifications was the most challenging part. It was hard for me to ensure that messages were delivered instantly.

5. User Profile Management

Brief Description: Users can manage their profiles, including updating personal information like name, email, and profile picture. The system also allows users to change their passwords securely, requiring the entry of the current password before a new one can be set.

Most Difficult Part: The most difficult part was implementing secure password management. Ensuring that the password change process was secure involved correctly handling user authentication, validating old passwords, and securely updating to the new password.

6. Genre Management

Brief Description: The genre management system allows admins or authorized users to add, edit, or delete genres. This ensures that the genre list remains up-to-date and relevant. Each book listing is categorized under a genre, which users can use to browse the platform's book offerings.

Most Difficult Part: The most challenging aspect was ensuring that genre changes (additions, deletions, or edits) were properly reflected across the platform. This required careful database management to maintain data consistency and ensure that books were always associated with the correct genre, even when changes were made.

Note: I haven't given any Instructions for the functionalities because they are self explanatory